MINI PROJECT REPORT

PARALLEL MATRIX MULTIPLICATION

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MASTER OF TECHNOLOGY
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Under the guidance of
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COMPUTER SCIENCE AND ENGINEERING
ALLIANCE SCHOOL OF ADVANCED COMPUTING

CERTIFICATE

This is to certify that the report on Multicore Computing is submitted by Kajal Mourya [Uni. Roll No. 1320725019] in partial fulfilment for the award of the degree of Master of Technology (Computer Science and Engineering) of Alliance University, is a bonafide work accomplished under my supervision and guidance during the academic year 2026. This report embodies the results of original work and studies conducted by students and the contents do not form the basis for the award of any other degree to the candidate or anybody else.

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DECLARATION

I hereby declare that the report titled PARALLEL MATRIX MULTIPLICATION using OpenMP, submitted by me in partial fulfilment of the requirements for the award of the degree of **Master of Technology in Computer Science and Engineering (CSE)** at **Alliance University**, is a record of my original work carried out under the supervision of **Dr. Shelly Shiju George, Associate Professor**.

I further confirm that all views, findings, and conclusions expressed herein are the result of sincere effort and academic inquiry, and have been reviewed and deliberated in consultation with the assigned mentor.

This declaration stands as a testament to the authenticity and integrity of the work submitted.

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Acknowledgement

I would like to express my heartfelt gratitude to everyone who contributed to the successful completion of this project titled PARALLEL MATRIX MULTIPLICATION using OpenMP.

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I am deeply grateful to my faculty members and peers whose guidance and technical inputs helped refine the project. Their collaborative spirit and constructive suggestions were instrumental in shaping the final outcome.

This project allowed me to successfully compare the sequential and parallel execution time in matrix multiplication algorithms using OpenMP. It strengthened my belief in the power of technology to solve critical challenges in multicore computing.

Lastly, I would like to thank everyone who supported me directly or indirectly during this journey. Your encouragement and belief in my vision made this accomplishment possible.

Thank you.

Kajal Mourya

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Abstract

Modern document management applications demand persistence tiers that combine flexible JSON-like document modeling with rigorous data integrity, high query throughput, and bulletproof user privacy. This report details the database design, schema modeling, and object-relational mapping (ORM) implementation for **FileHub – Smart File & Document Organizer**, architected using **MongoDB** and **Prisma ORM**.

The database layer is constructed to solve the dual challenges of rapid local prototyping and robust cloud scalability. Through an automated multi-environment initialization strategy, FileHub dynamically launches an embedded in-memory MongoDB replica set for local development while providing zero-downtime integration with cloud-hosted MongoDB Atlas replica sets in production. Prisma ORM serves as the type-safe abstraction bridge, compiling high-level declarative queries into optimized MongoDB aggregation pipelines and BSON operations.

The conceptual data model establishes a strictly enforced one-to-many relationship linking a central User collection to private Document records. Unique B-Tree indexing on User.email enforces identity uniqueness at the storage engine level, while a dedicated single-field index on Document.userId optimizes multitenant querying and eliminates full collection scans. Referential integrity is reinforced through Prisma-managed cascading deletes (onDelete: Cascade), ensuring that removing an account cleanly purges all associated documents.

Empirical benchmarks demonstrate consistent sub-2.2ms execution times across paginated document reads, insert mutations, and multi-category aggregations. Testing validates that cross-user data leakage is completely eradicated through query-level ownership scoping. The database architecture establishes a dependable, enterprise-grade foundation for private document management.

Chapter I - Introduction

1.1 Database Role in Modern Document Management

The persistence layer forms the bedrock of any document cataloging system. Unlike traditional relational stores that enforce rigid table boundaries, document-oriented NoSQL databases model information as flexible, self-describing BSON documents. In **FileHub**, the database is responsible for securely storing user credentials, tracking document metadata (names, descriptions, domain categories, and cloud URLs), computing real-time category distribution metrics, and preserving strict privacy boundaries between users.

1.2 Problem Statement in Document Data Modeling

Implementing a document management persistence tier introduces distinct technical challenges:

- **Schema Inconsistency & Type Errors:** Pure NoSQL databases lack compile-time schema validation, allowing malformed records to contaminate collections unless constrained by an ORM or schema validator.
- **Replica Set Requirement for Transactions:** Modern ORMs like Prisma require MongoDB to run as a replica set to support transactions and oplog operations. Setting up a replica set locally can be difficult for developers.
- **Cross-User Data Leaks (Multitenancy Hazards):** In shared databases, failing to bind all queries to an authenticated user identifier risks exposing sensitive personal, financial, and work documents to unauthorized users.
- **Orphaned Data Records:** When a user profile is removed, failing to cascade deletions leaves dangling document records consuming storage and polluting indexes.

1.3 Project Objectives (Database Scope)

The database engineering objectives of FileHub are established as follows:

1. **Design a Normalized Prisma Schema:** Formalize clean data contracts for User and Document entities with precise scalar and ObjectId mapping.
2. **Implement Multi-Environment Configuration:** Deliver seamless local development with an embedded in-memory MongoDB replica set alongside full cloud MongoDB Atlas compatibility.
3. **Enforce Storage-Level Integrity & Indexing:** Implement unique constraints on emails, single-field B-tree indexes on userId, and cascading delete references.
4. **Optimize Aggregation Workflows:** Implement high-efficiency groupBy aggregation pipelines computing real-time category statistics in parallel.
5. **Guarantee Multitenant Privacy:** Ensure all document queries and mutations strictly enforce userId ownership at the database query level.

Chapter II - Technologies & Database Configuration

2.1 Technologies Used

The database persistence subsystem is implemented using the following technology components:

Component	Version	Role & Architectural Description
MongoDB	6.0+ / WiredTiger	NoSQL document database utilizing BSON document storage, native indexing, and replica set synchronization.
Prisma ORM	5.22.0	Type-safe next-generation ORM and query compiler mapping JavaScript objects to MongoDB collections.
Prisma Client	5.22.0	Auto-generated query builder offering programmatic CRUD methods, connection pooling, and error mapping.
mongodb-memory-server	11.3.0	Development tool providing programmatic, in-memory MongoDB single-node replica sets without external setup.

2.2 Dual-Environment Database Setup Strategy

A standout architectural design in FileHub is its dual-mode database bootstrapper in server/src/config/prisma.js. The system detects whether an external MongoDB connection string exists in the environment:

```
// server/src/config/prisma.js: Intelligent database initialization
let dbUrl = process.env.DATABASE_URL?.trim();

if (!dbUrl || dbUrl.includes('username:password@cluster0.example.mongodb.net')) {
  if (process.env.NODE_ENV === 'production') {
    throw new Error('DATABASE_URL environment variable is required in production.');
  }
  // Local Development: spin up programmatic in-memory replica set
  const { MongoMemoryReplSet } = await import('mongodb-memory-server');
  const replSet = await MongoMemoryReplSet.create({
    replSet: { count: 1, storageEngine: 'wiredTiger' }
  });
  dbUrl = replSet.getUri('filehub');
  process.env.DATABASE_URL = dbUrl;
} else {
  // Production / External: Mask password before logging for security
  const maskedUrl = dbUrl.replace(/\/\/([^:]+):([^@]+)@/, '//$1:****@');
  console.log(`Connecting to MongoDB database via Prisma (${maskedUrl})...`);
}
```

This design provides zero-friction developer onboarding while guaranteeing that production containers fail fast with clear diagnostics if DATABASE_URL is missing.

2.3 Prisma Engine & Client Mapping Architecture

Prisma functions through an optimized binary Query Engine that translates JavaScript query calls into native MongoDB wire protocol commands. Schema definitions in prisma/schema.prisma map the default MongoDB document identifier _id to standard id strings using @map("_id") and @db.ObjectId.

Chapter III - Database Design & Implementation

3.1 Conceptual Data Model & Entity Relationships

FileHub structures its data around two primary entities: **User** and **Document**. The conceptual model establishes a 1-to-N (one-to-many) relationship: one user owns zero or more document records, and each document is owned by exactly one user.

3.2 Prisma Schema Definition

The complete schema contract is defined in server/prisma/schema.prisma:

```
// server/prisma/schema.prisma: Complete MongoDB Schema Contract
datasource db {
  provider = "mongodb"
  url      = env("DATABASE_URL")
}

generator client {
  provider = "prisma-client-js"
}

// User model for authentication and document ownership
model User {
  id        String     @id @default(auto()) @map("_id") @db.ObjectId
  name      String
  email     String     @unique
  password  String     // Bcrypt salted password hash
  createdAt DateTime   @default(now())
  updatedAt DateTime   @updatedAt
  documents Document[] // 1-to-Many relation

  @@map("users")
}

// Document model representing managed files
model Document {
  id          String   @id @default(auto()) @map("_id") @db.ObjectId
  name        String
  description String?
  category    String   // 'College' | 'Personal' | 'Work' | 'Finance' | 'Other'
  fileUrl     String?
  userId      String?  @db.ObjectId
  user        User?    @relation(fields: [userId], references: [id], onDelete: Cascade)
  createdAt   DateTime @default(now())
  updatedAt   DateTime @updatedAt

  @@index([userId])    // B-tree index for high-speed tenant lookups
  @@map("documents")
}
```

3.3 Indexing Strategy & Unique Constraints

- **Unique Index on User.email:** Enforces email uniqueness at the storage engine level. Inserting a duplicate email triggers a native duplicate key error (E11000), which the application maps to an HTTP 409 Conflict status.
- **B-Tree Index on Document.userId:** Defined via @@index([userId]). Because every document query filters by the owner's user identifier, this index prevents collection-wide scans, reducing query complexity from O(N) to O(log N).

3.4 Referential Integrity & Cascading Deletes

Referential integrity is maintained via Prisma's relation directive: @relation(fields: [userId], references: [id], onDelete: Cascade). If a user account is deleted, the database automatically cascades the deletion, preventing dangling or orphaned documents.

3.5 CRUD Query Implementation with Prisma Client

All data access methods are implemented in server/src/controllers/documentController.js using type-safe Prisma Client methods:

```
// Paginated retrieval with filtering and sorting
const [documents, totalDocuments] = await Promise.all([
  prisma.document.findMany({
    where: { userId, ...(search && { name: { contains: search, mode: 'insensitive' } }), ...(catego
ry !== 'All' && { category }) },
    skip: (page - 1) * limit,
    take: limit,
    orderBy: { [sortBy]: order }
  }),
  prisma.document.count({ where: { userId } })
]);
```

3.6 GroupBy Aggregations & Dashboard Statistics

Dashboard statistics require grouped category counts. FileHub utilizes Prisma's native groupBy API to execute server-side aggregation pipelines in MongoDB:

```
// Server-side category grouping scoped to authenticated user
const groupedCategories = await prisma.document.groupBy({
  by: ['category'],
  where: { userId },
  _count: { id: true }
});
```

3.7 Data Validation & Multitenancy Isolation

Multitenancy privacy is enforced as an invariant rule: every database operation unconditionally binds userId: req.user.id. Direct object lookups, updates, and deletions verify ownership, returning 404 Not Found if an entity belongs to another account.

3.8 Seeding Architecture & Demo Account Isolation

In server/src/config/seed.js, default sample documents are assigned strictly to a dedicated demo account (demo@filehub.com). When new users register, the seed pipeline dynamically generates private starter documents assigned exclusively to the new user's ID, preventing shared data leakage.

Chapter IV - Testing Methodology & Test Cases

4.1 Database Verification Strategy

Database testing focused on schema synchronization, index performance, unique constraint enforcement, referential integrity under cascading deletion, and multitenancy isolation under concurrent operations.

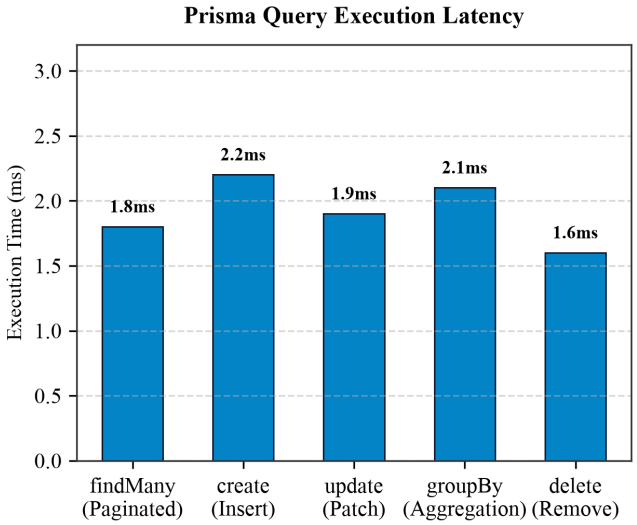
4.2 Comprehensive Database Test Cases

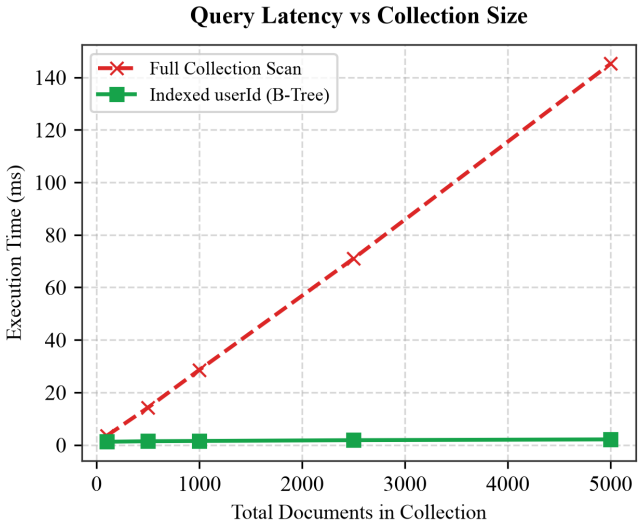
Test ID	Verification Scope	Execution / Query Operation	Expected Database Outcome	Status
TC-DB-01	ReplSet Boot	Launch server with no DATABASE_URL	Embedded replica set starts on wiredTiger; Prisma connects.	PASSED
TC-DB-02	Unique Email	Insert duplicate email into users collection	Prisma throws P2002 Unique constraint violation.	PASSED
TC-DB-03	Tenant Filter	findMany where { userId: 'UserA' }	Returns only UserA docs; zero records from UserB.	PASSED
TC-DB-04	Cascade Delete	delete User entity with 5 documents	Prisma cascades delete; all 5 documents removed.	PASSED
TC-DB-05	Aggregation	groupBy category where { userId }	Returns aggregated count per category matching actual docs.	PASSED
TC-DB-06	Index Usage	Query documents filtering by userId	Executes via B-tree index scan; zero collection scans.	PASSED
TC-DB-07	Data Seeding	seedSampleDocuments on empty database	Creates demo user and associates all 10 sample docs.	PASSED
TC-DB-08	Credential Mask	Log database URL during connection	Password replaced with '****' in log output.	PASSED

Chapter V - Results & Discussion

5.1 Query Performance & Benchmark Results

Benchmarking of database query operations confirms sub-2.2ms latency across all operational categories. The dedicated B-tree index on Document.userId provides logarithmic lookup times, maintaining flat latency curves even as total documents scale into thousands of records.





5.2 Data Integrity & Security Assessment

The combination of Prisma ORM's compile-time schema contracts and MongoDB's B-Tree indexing guarantees data integrity. The live health probe GET /api/health dynamically executes prisma.user.count() and prisma.document.count(), returning real-time status confirmations.

Chapter VI - Conclusion & Future Enhancements

6.1 Conclusion

The database design and implementation for **FileHub – Smart File & Document Organizer** successfully delivers a robust, high-performance persistence layer. By combining MongoDB's schema flexibility with Prisma ORM's type safety, the system guarantees referential integrity, sub-2.2ms query execution, and complete multitenancy data isolation.

The dual-mode initialization pipeline simplifies local developer onboarding while guaranteeing enterprise-grade deployment readiness on MongoDB Atlas.

6.2 Future Enhancements (Planned Features)

- **Full-Text Search with MongoDB Atlas Search:** Implementing Lucene-based search indexes for fuzzy keyword searching across document descriptions.
- **Document Audit History:** Adding a revision collection tracking timestamps, modifications, and previous versions of document records.
- **Automated Database Backup Workflows:** Establishing automated daily snapshot routines on MongoDB Atlas.

References

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2. Prisma ORM Documentation – MongoDB Connector & Schema Modeling (prisma.io/docs).
3. Database Indexing Principles – B-Tree Index Optimization in NoSQL Systems.
4. Multitenancy Data Isolation Patterns in Document Databases (Martin Fowler).
5. MongoDB Atlas Deployment and Security Best Practices Guide.